

HTML Tags and CSS Selectors Reference Manual

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HTML Structure Tags

Document Structure

```
<!DOCTYPE html>           <!-- Declares HTML5 document type -->
<html lang="en">           <!-- Root element with language attribute -
->

  <head>                     <!-- Contains metadata -->
  <title>Page Title</title> <!-- Page title (appears in browser tab) -
->

  <meta charset="UTF-8">    <!-- Character encoding -->
  <link rel="stylesheet" href="styles.css"> <!-- External CSS -->
  <script src="script.js"></script>        <!-- External JavaScript -
->

  </head>
  <body>                     <!-- Contains visible page content -->
  </body>
</html>
```

Basic Structure Elements

- `<html>` - Root element of HTML document
 - `<head>` - Contains metadata (not visible on page)
 - `<title>` - Document title (browser tab/window title)
 - `<body>` - Contains all visible content
 - `<meta>` - Metadata (charset, viewport, description, etc.)
 - `<link>` - Links to external resources (CSS, fonts, etc.)
 - `<script>` - JavaScript code or external script references
 - `<style>` - Internal CSS styles
-

HTML Content Tags

Headings

<code><h1>Main Heading</h1></code>	<code><!-- Most important heading --></code>
<code><h2>Section Heading</h2></code>	<code><!-- Secondary heading --></code>
<code><h3>Subsection</h3></code>	<code><!-- Third level heading --></code>
<code><h4>Minor Heading</h4></code>	<code><!-- Fourth level heading --></code>
<code><h5>Small Heading</h5></code>	<code><!-- Fifth level heading --></code>
<code><h6>Smallest Heading</h6></code>	<code><!-- Least important heading --></code>

Text Content

<code><p>Paragraph text content</p></code>	<code><!-- Paragraph --></code>
<code>
</code>	<code><!-- Line break</code>
<code>(self-closing) --></code>	
<code><hr></code>	<code><!-- Horizontal</code>
<code>rule (self-closing) --></code>	
<code>Bold/Important text</code>	<code><!-- Strong</code>
<code>emphasis (bold) --></code>	
<code>Bold text</code>	<code><!-- Bold (visual</code>
<code>only) --></code>	
<code>Emphasized text</code>	<code><!-- Emphasis</code>
<code>(italic) --></code>	
<code><i>Italic text</i></code>	<code><!-- Italic (visual</code>
<code>only) --></code>	
<code><u>Underlined text</u></code>	<code><!-- Underlined text</code>
<code>--></code>	
<code>Inline container</code>	<code><!-- Generic inline</code>
<code>container --></code>	
<code><div>Block container</div></code>	<code><!-- Generic block</code>
<code>container --></code>	

Links and Images

<code>Link text</code>	<code><!-- External link -</code>
<code>--></code>	
<code>Internal link</code>	<code><!-- Internal link -</code>
<code>--></code>	
<code>Anchor link</code>	<code><!-- Link to page</code>
<code>section --></code>	
<code>Email</code>	<code><!-- Email link --></code>
<code></code>	<code><!-- Image (self-</code>
<code>closing) --></code>	
<code></code>	<code><!--</code>
<code>- Image with dimensions --></code>	

Lists

```
<!-- Unordered List (bullets) -->
<ul>
  <li>First item</li>
  <li>Second item</li>
  <li>Third item</li>
</ul>
```

```

<!-- Ordered List (numbers) -->
<ol>
  <li>First step</li>
  <li>Second step</li>
  <li>Third step</li>
</ol>

<!-- Description List -->
<dl>
  <dt>Term 1</dt>
  <dd>Definition of term 1</dd>
  <dt>Term 2</dt>
  <dd>Definition of term 2</dd>
</dl>

```

Tables

```

<table>
  <caption>Table Caption</caption>
  <thead>
    <tr>
      <th>Header 1</th>
      <th>Header 2</th>
      <th>Header 3</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Data 1</td>
      <td>Data 2</td>
      <td>Data 3</td>
    </tr>
    <tr>
      <td colspan="2">Spans 2 columns</td>
      <td>Data 6</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td>Footer 1</td>
      <td>Footer 2</td>
      <td>Footer 3</td>
    </tr>
  </tfoot>
</table>

```

Table Attributes

- border="1" - Table border width
 - cellpadding="5" - Space inside cells
 - cellspacing="5" - Space between cells
 - colspan="2" - Cell spans multiple columns
 - rowspan="2" - Cell spans multiple rows
 - align="center" - Horizontal alignment
 - valign="top" - Vertical alignment
-

HTML Form Elements

Form Structure

```
<form action="submit.php" method="post" target="_blank">
  <!-- Form elements go here -->
</form>
```

Input Types

```
<!-- HTML4 Input Types -->
<input type="text" name="username" placeholder="Enter username">
<input type="password" name="password" placeholder="Password">
<input type="checkbox" name="agree" value="yes"> I agree
<input type="radio" name="gender" value="male"> Male
<input type="radio" name="gender" value="female"> Female
<input type="submit" value="Submit">
<input type="reset" value="Reset">
<input type="button" value="Click Me" onclick="alert('Clicked!')">
<input type="hidden" name="userId" value="123">
<input type="file" name="upload" accept=".jpg,.png,.pdf">

<!-- HTML5 Input Types -->
<input type="email" name="email" placeholder="email@example.com">
<input type="url" name="website" placeholder="https://example.com">
<input type="tel" name="phone" placeholder="123-456-7890">
<input type="number" name="age" min="1" max="100" step="1">
<input type="range" name="volume" min="0" max="100" value="50">
<input type="date" name="birthday">
<input type="time" name="appointment">
<input type="datetime-local" name="meeting">
<input type="month" name="expiry">
<input type="week" name="vacation">
<input type="color" name="theme">
<input type="search" name="query" placeholder="Search...">
```

Other Form Elements

```
<textarea name="comments" rows="4" cols="50" placeholder="Enter
comments"></textarea>

<select name="country">
  <option value="">Choose country</option>
  <option value="us">United States</option>
  <option value="uk" selected>United Kingdom</option>
  <option value="ca">Canada</option>
</select>

<button type="submit">Submit Form</button>
<button type="reset">Reset Form</button>
<button type="button" onclick="doSomething()">Click Me</button>

<fieldset>
  <legend>Personal Information</legend>
  <!-- Related form elements -->
</fieldset>
```

```
<label for="username">Username:</label>
<input type="text" id="username" name="username">
```

HTML5 Form Attributes

```
<input type="text"
      name="username"
      placeholder="Enter username"
      required
      autofocus
      autocomplete="off"
      pattern="[A-Za-z0-9]{3,20}"
      minlength="3"
      maxlength="20">

<input type="email"
      name="email"
      placeholder="email@example.com"
      required
      multiple>

<input type="file"
      name="documents"
      accept=".pdf,.doc,.docx"
      multiple>
```

HTML5 Semantic Elements

Layout Semantics

```
<header>
  <nav>
    <ul>
      <li><a href="#home">Home</a></li>
      <li><a href="#about">About</a></li>
      <li><a href="#contact">Contact</a></li>
    </ul>
  </nav>
</header>

<main>
  <section>
    <h1>Main Content Section</h1>
    <article>
      <h2>Article Title</h2>
      <p>Article content...</p>
    </article>

    <aside>
      <h3>Related Information</h3>
      <p>Sidebar content...</p>
    </aside>
  </section>
</main>

<footer>
```

```
<p>&copy; 2024 Website Name. All rights reserved.</p>
</footer>
```

Content Semantics

```
<figure>
  
  <figcaption>Sales data for Q4 2024</figcaption>
</figure>

<details>
  <summary>Click to expand</summary>
  <p>Hidden content that can be toggled</p>
</details>

<mark>Highlighted text</mark>
<time datetimes="2024-01-15">January 15, 2024</time>

<address>
  Contact us at: <a
href="mailto:info@example.com">info@example.com</a><br>
  123 Main Street, City, State 12345
</address>
```

Multimedia Elements

```
<!-- Audio -->
<audio controls>
  <source src="audio.mp3" type="audio/mpeg">
  <source src="audio.ogg" type="audio/ogg">
  Your browser does not support audio.
</audio>

<!-- Video -->
<video controls width="640" height="480">
  <source src="video.mp4" type="video/mp4">
  <source src="video.webm" type="video/webm">
  Your browser does not support video.
</video>
```

HTML Attributes

Global Attributes (Available on all elements)

<pre><div id="unique-identifier"></pre>	<pre><!-- Unique identifier --></pre>
<pre><div class="css-class another"></pre>	<pre><!-- CSS classes (space-separated) --></pre>
<pre><div style="color: red;"></pre>	<pre><!-- Inline CSS styles --></pre>
<pre><div title="Tooltip text"></pre>	<pre><!-- Tooltip on hover --></pre>
<pre><div data-custom="value"></pre>	<pre><!-- Custom data attributes --></pre>
<pre><div lang="en"></pre>	<pre><!-- Language --></pre>
<pre><div hidden></pre>	<pre><!-- Hide element --></pre>
<pre><div contenteditable="true"></pre>	<pre><!-- Make element editable --></pre>
<pre><div draggable="true"></pre>	<pre><!-- Make element draggable --></pre>
<pre><div tabindex="1"></pre>	<pre><!-- Tab order for keyboard navigation --></pre>

Common Specific Attributes

```
<!-- Links -->
<a href="url" target="_blank" rel="noopener">

<!-- Images -->


<!-- Forms -->
<input type="text" name="field" value="default" placeholder="hint"
required disabled readonly>
<form action="submit.php" method="post" enctype="multipart/form-
data">

<!-- Tables -->
<td colspan="2" rowspan="3" align="center" valign="top">

<!-- Media -->
<audio src="audio.mp3" controls autoplay loop muted preload="auto">
<video src="video.mp4" controls autoplay loop muted
poster="thumbnail.jpg">
```

CSS Selectors

Basic Selectors

```
/* Element Selector */
p { color: blue; }           /* All <p> elements */
h1 { font-size: 24px; }     /* All <h1> elements */

/* Class Selector */
.highlight { background: yellow; } /* Elements with
class="highlight" */
.container { width: 100%; }       /* Elements with
class="container" */

/* ID Selector */
#header { background: navy; }     /* Element with id="header"
*/
#sidebar { width: 200px; }        /* Element with id="sidebar"
*/

/* Universal Selector */
* { margin: 0; padding: 0; }     /* All elements */
```

Compound Selectors

```
/* Multiple Classes */
.large.highlight { font-size: 20px; } /* Elements with BOTH classes
*/

/* Element with Class */
p.important { font-weight: bold; }    /* <p> elements with
class="important" */
div.container { max-width: 1200px; } /* <div> elements with
```

```

class="container" */

/* Element with ID */
div#main { background: white; } /* <div> element with
id="main" */

```

Grouping Selectors

```

/* Multiple Selectors (comma-separated) */
h1, h2, h3 { color: navy; } /* All h1, h2, and h3 elements
*/

.header, .footer { background: gray; } /* Elements with either class
*/

#nav, #sidebar { border: 1px solid; } /* Elements with either ID */

```

Descendant Selectors

```

/* Descendant Selector (space) */
.container p { margin: 10px; } /* <p> elements inside
.container */
#sidebar ul { list-style: none; } /* <ul> elements inside
#sidebar */

/* Child Selector (>) */
.menu > li { display: inline; } /* Direct <li> children of
.menu */
#content > h2 { border-bottom: 1px; } /* Direct <h2> children of
#content */

/* Adjacent Sibling (+) */
h1 + p { margin-top: 0; } /* <p> immediately after <h1>
*/

/* General Sibling (~) */
h2 ~ p { color: gray; } /* All <p> siblings after <h2>
*/

```

Attribute Selectors

```

/* Has Attribute */
[title] { cursor: help; } /* Elements with title
attribute */
[data-role] { position: relative; } /* Elements with data-role
attribute */

/* Exact Attribute Value */
[type="text"] { border: 1px solid; } /* Elements with type="text"
*/

[class="highlight"] { background: yellow; } /* Exact class match */

/* Attribute Contains Word */
[class~="nav"] { display: flex; } /* Class contains "nav" as
whole word */

/* Attribute Starts With */
[href^="https"] { color: green; } /* href starts with "https" */
[class^="btn"] { padding: 10px; } /* class starts with "btn" */

```

```

/* Attribute Ends With */
[href$=".pdf"] { font-weight: bold; } /* href ends with ".pdf" */
[src$=".jpg"] { border-radius: 5px; } /* src ends with ".jpg" */

/* Attribute Contains Substring */
[href*="example"] { text-decoration: underline; } /* href contains
"example" */

```

Pseudo-Class Selectors

```

/* Link States */
a:link { color: blue; } /* Unvisited links */
a:visited { color: purple; } /* Visited links */
a:hover { color: red; } /* Mouse hover */
a:active { color: orange; } /* Being clicked */
a:focus { outline: 2px solid; } /* Keyboard focus */

/* Structure Pseudo-Classes */
:first-child { margin-top: 0; } /* First child element */
:last-child { margin-bottom: 0; } /* Last child element */
:nth-child(2n) { background: #f0f0f0; } /* Even children */
:nth-child(odd) { background: white; } /* Odd children */
:nth-child(3n+1) { color: red; } /* Every 3rd starting from 1st */
*/

/* Type-based Structure */
:first-of-type { font-weight: bold; } /* First element of its type */
*/
:last-of-type { border-bottom: none; } /* Last element of its type */
*/
:nth-of-type(2) { color: blue; } /* Second element of its type */
*/
:only-child { text-align: center; } /* Only child element */
:only-of-type { font-style: italic; } /* Only element of its type */

/* Form States */
input:focus { border-color: blue; } /* Input has focus */
input:disabled { opacity: 0.5; } /* Disabled input */
input:checked { transform: scale(1.1); } /* Checked checkbox/radio */
*/
input:required { border-left: 3px solid red; } /* Required field */
input:valid { border-color: green; } /* Valid input */
input:invalid { border-color: red; } /* Invalid input */

/* Other States */
:empty { display: none; } /* Empty elements */
:not(.special) { color: black; } /* Elements without .special
class */

```

Pseudo-Element Selectors

```

/* Content Pseudo-Elements */
::before { content: "★ "; } /* Insert content before
element */
::after { content: " →"; } /* Insert content after
element */

/* Text Pseudo-Elements */
::first-letter { font-size: 2em; } /* First letter of element */

```

```
::first-line { font-weight: bold; } /* First line of element */

/* Selection */
::selection { background: yellow; } /* Selected text */
```

CSS Properties

Text Properties

```
.text-styles {
  color: #333333; /* Text color */
  font-family: Arial, sans-serif; /* Font family */
  font-size: 16px; /* Font size */
  font-weight: bold; /* Font weight (normal, bold,
100-900) */
  font-style: italic; /* Font style (normal, italic,
oblique) */
  text-align: center; /* Text alignment (left,
center, right, justify) */
  text-decoration: underline; /* Text decoration (none,
underline, line-through) */
  text-transform: uppercase; /* Text transform (none,
uppercase, lowercase, capitalize) */
  line-height: 1.5; /* Line height */
  letter-spacing: 2px; /* Space between letters */
  word-spacing: 5px; /* Space between words */
  text-indent: 20px; /* First line indentation */
  white-space: nowrap; /* White space handling */
}
```

Background Properties

```
.background-styles {
  background-color: #f0f0f0; /* Background color */
  background-image: url('image.jpg'); /* Background image */
  background-repeat: no-repeat; /* Image repeat (repeat, no-
repeat, repeat-x, repeat-y) */
  background-position: center top; /* Image position */
  background-size: cover; /* Image size (auto, contain,
cover, 100px 50px) */
  background-attachment: fixed; /* Scroll behavior (scroll,
fixed, local) */

  /* Shorthand */
  background: #fff url('bg.jpg') no-repeat center/cover;
}
```

Border Properties

```
.border-styles {
  border-width: 2px; /* Border width */
  border-style: solid; /* Border style (solid, dashed,
dotted, double, etc.) */
  border-color: #333; /* Border color */
  border-radius: 10px; /* Rounded corners */
}
```

```

/* Individual sides */
border-top: 1px solid red;
border-right: 2px dashed blue;
border-bottom: 3px dotted green;
border-left: 4px double orange;

/* Individual corner radius */
border-top-left-radius: 5px;
border-top-right-radius: 10px;
border-bottom-right-radius: 15px;
border-bottom-left-radius: 20px;

/* Shorthand */
border: 2px solid #333;
}

```

Spacing Properties

```

.spacing {
  margin: 20px;                /* All sides */
  margin: 10px 20px;          /* Top/bottom, left/right */
  margin: 10px 15px 20px 25px; /* Top, right, bottom, left */

  padding: 15px;              /* All sides */
  padding: 10px 20px;         /* Top/bottom, left/right */
  padding: 5px 10px 15px 20px; /* Top, right, bottom, left */

  /* Individual sides */
  margin-top: 10px;
  margin-right: 20px;
  margin-bottom: 15px;
  margin-left: 5px;

  padding-top: 8px;
  padding-right: 12px;
  padding-bottom: 16px;
  padding-left: 4px;
}

```

Display and Visibility

```

.display-options {
  display: block;           /* Block element */
  display: inline;          /* Inline element */
  display: inline-block;    /* Inline-block element */
  display: flex;            /* Flexbox container */
  display: grid;            /* Grid container */
  display: none;            /* Hide element */

  visibility: hidden;       /* Hide but keep space */
  visibility: visible;      /* Show element */

  opacity: 0.5;             /* Transparency (0-1) */
}

```

CSS Box Model

Box Model Properties

```
.box-model {  
  /* Content dimensions */  
  width: 300px;  
  height: 200px;  
  min-width: 200px;  
  max-width: 400px;  
  min-height: 100px;  
  max-height: 300px;  
  
  /* Box sizing */  
  box-sizing: border-box;           /* Include padding and border in  
width/height */  
  box-sizing: content-box;         /* Default: only content in  
width/height */  
  
  /* Padding (inside border) */  
  padding: 20px;  
  
  /* Border */  
  border: 2px solid #333;  
  
  /* Margin (outside border) */  
  margin: 15px;  
  
  /* Outline (outside margin, doesn't affect layout) */  
  outline: 1px dashed red;  
  outline-offset: 5px;  
}
```

CSS Positioning

Position Property

```
.positioning {  
  position: static;           /* Default: normal document  
flow */  
  position: relative;         /* Relative to normal position  
*/  
  position: absolute;         /* Relative to positioned  
ancestor */  
  position: fixed;            /* Relative to viewport */  
  position: sticky;           /* Sticky positioning */  
  
  /* Position coordinates (used with relative, absolute, fixed,  
sticky) */  
  top: 10px;  
  right: 20px;  
  bottom: 30px;  
  left: 40px;  
  
  /* Z-index (stacking order) */  
  z-index: 100;  
}
```

Float and Clear

```
.floating {  
  float: left;           /* Float left */  
  float: right;          /* Float right */  
  float: none;           /* No float */  
  
  clear: left;           /* Clear left floats */  
  clear: right;          /* Clear right floats */  
  clear: both;           /* Clear all floats */  
  clear: none;           /* No clear */  
}
```

Overflow

```
.overflow-control {  
  overflow: visible;      /* Content overflows (default) */  
  /*  
  overflow: hidden;       /* Hide overflowing content */  
  overflow: scroll;        /* Always show scrollbars */  
  overflow: auto;         /* Show scrollbars when needed */  
  /*  
  overflow-x: hidden;     /* Horizontal overflow */  
  overflow-y: scroll;      /* Vertical overflow */  
}
```

Media Queries and Responsive Design

```
/* Mobile First Approach */  
.container {  
  width: 100%;  
  padding: 10px;  
}  
  
/* Tablet and up */  
@media screen and (min-width: 768px) {  
  .container {  
    width: 750px;  
    margin: 0 auto;  
  }  
}  
  
/* Desktop and up */  
@media screen and (min-width: 1024px) {  
  .container {  
    width: 1200px;  
  }  
}  
  
/* Print styles */  
@media print {  
  .no-print {  
    display: none;  
  }  
}
```

Common CSS Patterns

Center Alignment

```
/* Horizontal centering */
.center-horizontal {
  margin: 0 auto;           /* Block element */
  text-align: center;       /* Inline content */
}

/* Vertical centering */
.center-vertical {
  display: flex;
  align-items: center;      /* Flex method */

  /* OR */
  position: relative;
  top: 50%;
  transform: translateY(-50%); /* Transform method */
}

/* Both horizontal and vertical */
.center-both {
  display: flex;
  justify-content: center;
  align-items: center;
}
```

Clearfix (for floated elements)

```
.clearfix::after {
  content: "";
  display: table;
  clear: both;
}
```

Button Styling

```
.button {
  display: inline-block;
  padding: 10px 20px;
  background: #007bff;
  color: white;
  text-decoration: none;
  border-radius: 4px;
  border: none;
  cursor: pointer;
  transition: background-color 0.3s;
}

.button:hover {
  background: #0056b3;
}

.button:active {
  transform: translateY(1px);
}
```

This reference manual covers the essential HTML tags and CSS selectors from your PDF, organized in a practical way for quick reference and learning.